

Amazon Reviews Sentiment Analysis

Introduction

In today's digital era, customer reviews play a crucial role in shaping product success on e-commerce platforms. This project focuses on analyzing Amazon customer reviews to extract meaningful insights about user satisfaction, product quality, and overall sentiment.

The main objective of this project is to understand customer behavior by analyzing ratings and textual reviews using sentiment analysis techniques. By combining data processing with interactive dashboards, this project helps in identifying trends, patterns, and areas of improvement for businesses.

Dataset Description

The dataset used in this project is the Amazon Customer Reviews Dataset, which contains detailed information about customer feedback.

Key Features of Dataset:

- Review Text (Customer feedback)
- Rating (1 to 5 stars)
- Review Date
- Reviewer Information

Data Preprocessing Steps:

- Removed missing and null values
 - Converted data types (e.g., date format correction)
 - Cleaned review text for analysis
 - Handled inconsistent values (e.g., "1 review", "23 reviews")
 - Filled missing values where necessary (e.g., NA for empty fields)
-

Tools & Technologies Used

- Power BI → Dashboard creation & visualization

- Python → Data cleaning and sentiment analysis
 - Pandas → Data manipulation
 - NLTK & TextBlob → Sentiment analysis
-

Power BI Visualizations & Insights

1) Cards (Overview Section)

This section displays key metrics such as:

- Total Reviews (~21K+)
- Average Rating
- Average Sentiment Score

Insight:

These KPIs provide a quick summary of overall dataset performance. Despite having a large number of reviews, the sentiment is mixed, indicating inconsistent customer experience.

2) Sentiment Distribution (Donut Chart)

This visualization shows the proportion of:

- Positive Reviews (~48%)
- Negative Reviews (~44%)
- Neutral Reviews

Insight:

Although positive reviews are slightly higher, the negative reviews are also significantly high. This suggests that there is a strong need for product or service improvement.

3) Sentiment Over Time (Line Chart)

This chart tracks how sentiment changes over time based on review dates.

Insight:

The sentiment shows a slight decline over time, which may indicate decreasing customer satisfaction or quality issues in recent periods.

4) Rating Distribution (Pie Chart)

Displays count of ratings from 1-star to 5-star.

Insight:

A large number of users have given **1-star ratings**, which is a strong indicator of dissatisfaction. This highlights major issues with the product or service.

5) Top Reviewers (Bar Chart)

Shows users who have given the highest number of reviews.

Insight:

Some users are highly active and contribute multiple reviews. These users can be valuable for identifying consistent feedback patterns.

6) Sentiment vs Rating (Table Visualization)

Displays relationships between ratings and sentiment categories.

Insight:

- High ratings (4–5 stars) → Mostly Positive
- Low ratings (1–2 stars) → Mostly Negative

This confirms the expected correlation between rating and sentiment accuracy.

7) Slicers / Filters

Used filters include:

- Review Date
- Rating
- Sentiment

Insight:

Slicers allow dynamic analysis, enabling users to explore data based on specific time periods or rating categories.

Key Insights Summary

- Total ~21K+ reviews were analyzed
 - Majority of reviews are positive (~48%)
 - Negative reviews are also high (~44%), indicating improvement scope
 - Most users gave 1-star ratings → strong dissatisfaction signal
 - Sentiment slightly declined over time
 - Higher ratings align strongly with positive sentiment
-

Conclusion

This project demonstrates the power of combining data analysis, sentiment analysis, and visualization to understand customer feedback effectively. The findings reveal that while many users are satisfied, a significant portion of negative reviews highlights critical issues that need attention.

Overall, this dashboard provides a comprehensive view of customer sentiment and serves as a valuable tool for business analysis.

Business Insights from Sentiment Analysis

Company Areas for Improvement

The sentiment analysis reveals several key areas where the company needs to focus on improvement.

- A significant number of negative reviews highlight issues related to customer service, indicating that customers are not satisfied with the support they receive.
- Improving response quality and ensuring polite, helpful interactions can greatly enhance user experience.
- Delivery time is another major concern, as many customers have reported delays in receiving their orders.
- The company should work on optimizing its logistics and supply chain processes to ensure faster and more reliable delivery.
- Order management issues, such as incorrect or missing products, also contribute to customer dissatisfaction. Implementing better quality checks and tracking systems can help reduce such errors.
- Additionally, customers have reported account-related problems, including login issues and account access difficulties, which need to be addressed to ensure a smooth user experience.

Lastly, slow response times from support teams further frustrate customers. Reducing wait times and providing quicker resolutions can significantly improve overall satisfaction.

Customer Expectations

Based on the analysis, it is clear that customers expect fast and reliable delivery services, as timely order fulfillment plays a crucial role in their overall experience. They also value efficient and responsive customer support, where their issues are addressed quickly and effectively.

Customers prefer receiving correct and safely delivered products without any damage or discrepancies. A smooth and hassle-free account experience, including easy login and navigation, is also an important expectation.

Furthermore, customers seek trustworthy and consistent service, especially those using premium services like Prime, where higher expectations are associated with faster delivery and better support. Meeting these expectations can help build customer loyalty and improve brand reputation.